

FUSE: Finding File Upload Bugs via Penetration Testing

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An Unrestricted File Upload (UFU) vulnerability is a critical security threat that enables an adversary to upload her choice of a forged file to a target web server. This bug evolves into an Unrestricted Executable File Upload (UEFU) vulnerability when the adversary is able to conduct remote code execution of the uploaded file via triggering its URL. We design and implement FUSE, the first penetration testing tool designed to discover UFU and UEFU vulnerabilities in server-side PHP web applications. The goal of FUSE is to generate upload requests; each request becomes an exploit payload that triggers a UFU or UEFU vulnerability. However, this approach entails two technical challenges: (1) it should generate an upload request that bypasses all content-filtering checks present in a target web application; and (2) it should preserve the execution semantic of the resulting uploaded file. We address these technical challenges by mutating standard upload requests with carefully designed mutation operations that enable the bypassing of content-filtering checks and do not tamper with the execution of uploaded files. FUSE discovered 30 previously unreported UEFU vulnerabilities, including 15 CVEs from 33 real-world web applications, thereby demonstrating its efficacy in finding code execution bugs via file uploads.

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